

Fajri Ilham Mughni, S.T.

Jakarta, Indonesia | [Phone-Number](#) | [E-mail](#) | [Github](#) | [Medium](#)

Overview

A highly motivated individual with a strong background in data management, organizational leadership, and scholarship achievements. Proven communication, analytical, and teamwork skills through roles in student councils and national programs. Passionate about continuous learning, community development, and making meaningful contributions to both technical and social initiatives.

Educations

- Bandung Institute Of Technology (ITB) June 2018 – April 2023
Bachelor of Engineering : GPA 3.26/4.00 (154 Credits)
(Sarjana Teknik Geologi)
“Geological Engineering”
Thesis Title : Evaluasi Geologi Lingkungan Untuk Pengembangan Wilayah Kecamatan Sentolo Dan Sekitarnya, Kabupaten Kulon Progo, Provinsi Daerah Istimewa Yogyakarta. (Environmental Geology)
Competition : Awarded a Verbal Commendation at Paragon Innovation Circle (2021) for proposing a multi-stakeholder innovation on green industry implementation in Indonesia, and selected as a Top 10 Finalist in the Business Idea & Implementation Competition at PEC EXPO 1.0, Universitas Airlangga (2020).
Scholarship : Recipient of national and institutional scholarships recognizing academic excellence and leadership, including Beasiswa Luar Biasa ITB Batch 9, Purnomo Yusgiantoro Center (PYC), Dato’ Low Tuck Scholarship
Leadership & Activity :
 - Led well-being programs as Spiritual Department Manager of HMTG "GEA" ITB (2021), advocating mental health through three initiatives and achieving 100% KPIs; previously headed the Fundraising Department of the FITB Student Council (2018), creating programs to generate income and support all faculty student-wide activities.
 - Selected for the Kampus Merdeka Program at the Design Ethnography Laboratory (DE Lab), FSRD ITB, for a short-course preparation program and served as Assistant Researcher in multidisciplinary exploration and design-based social mapping in Larantuka, East Nusa Tenggara.
 - Participated in Kelas Akal Budi by the SabangMerauke Foundation (2020), focusing on empathy-building and cross-cultural education in response to the Covid-19.
 - Joined the short-incubation program StudentCatalyst Jakarta Chapter 2 (2018), aimed at accelerating and promoting youth involvement in social projects, entrepreneurship, and competition preparation.
 - Contributing writer at SuaraGEA Zine 2021, producing popular articles on social-geological perspectives.
- Short-term professional course Dibimbing.id November 2024 – May 2025
Data Science and Analysis : Score 77.61/100.00 (21 Weekly Topic)
Final Project Title :
 - Crisis and Analysis: Data-Driven Insights into the 2008 US Economic Crisis through Journal and Academic Publication Impact
 - What Do You Like to Eat? A Personalized Indonesian Food Recommendation System

Expertise

Technical Skills	: Python • SQL • HTML/CSS • GitHub • ETL Processes • Web Scraping • Power BI • Google Data Studio • Microsoft Office • Google Workspace • Adobe Suite • CorelDRAW • Cloud & On-Premise Storage Maintenance • Website Development
Geospatial & Mapping Tools	: QGIS • Google Earth Engine (GEE) • USGS Earth Explorer • Migas Data Repository (MDR) & other Indonesian Earth-Science Government Databases • ILWIS • Lithological Mapping • Environmental (Geo-hazard & Natural Resources) Mapping • Geospatial Analysis (Satellite Imagery – Landsat, etc.)
Data & Analysis Skills	: RFM Analysis • A/B Testing • SQL Querying • Exploratory Data Analysis (EDA) • Data Visualization • Statistical Analysis • Hypothesis Testing • Linear Regression • Climate Data Analysis for Hydrogeological Estimation (FJ. Mock Method) • Sustainability-Oriented Decision Making using Machine Learning & AI (e.g., SMCE)
Domain Expertise	: Environmental Analysis (AMDAL) • Geological-Spatial Analysis • Social Impact Analysis • Climate-Groundwater Analysis • Analytical Thinking for Spatial Planning (AHP-SMCE) • Multidisciplinary Business Understanding • Research & Reporting (Earth and Social Sciences) • Multidisciplinary Research Collaboration
Attachment	: Project Based Portfolio

Relevant Experience

- Husky-CNOOC Madura Limited (HCML)
Database Management Officer under a Third-Party Contract with PT Sigma Cipta Utama
February 2024 – July 2025

Task and Responsibility :
 - Supported Exploration Department operations by collecting over 100 data points from internal archives and the MDR of the Ministry of Energy and Mineral Resources. Consolidated and structured over 5,000 print and digital records into a centralized database.
 - Managed cross-storage data (on-premises and/or cloud), including planning, migration, documentation, search system activation, and file/folder management using AI and Machine Learning automation.
 - Maintained both physical and digital storage media of collected data, with monthly documentation updates.
 - Completed certifications: SNI ISO 30301, Geospatial Data Analysis Using Python, VBA-Macro, and Data Science & Analysis.
- Resilient Development Initiative (RDI Indonesia)
Disaster Preparedness and Community Resilience Intern
February 2023 – May 2023

Task and Responsibility :
 - Contributed to the Baseline Report for the SinerGi II Project, designing data visualizations (line and bar charts) of survey data, surpassing 100% target achievement
 - Integrated primary interview data and secondary datasets (Google Spreadsheet, Kobo Toolbox) into the SinerGi II framework, fully meeting project goals
 - Assisted in primary data collection using FGD and KII methods; observed enumerators conducting surveys for WVI's evaluation, covering 10+ key informants

3. Pusat Air Tanah & Geologi Tata Lingkungan ESDM March 2022 – July 2022
Environmental Geologist Intern
Task and Responsibility : • Assisted in environmental geology mapping for the “Integrated Environmental Geology of DIY Province” project, contributing to my bachelor thesis and presenting findings to PATGTL. Developed land change data for Yogyakarta by integrating environmental geology maps, satellite imagery, and historical land use data, utilizing over five geospatial tools to create land change maps from 1997 to 2023. Designed the “Evaluation for Development City” project in Kulon Progo Regency by combining hydrogeology, geo-hazard, accessibility, natural resources, and lithology data with the Analytic Hierarchy Process (AHP), culminating in a successful presentation after three months of research.

Project Experience

1. Leave No One Behind (LNOB) Study 2023 by United Nations (UN) Indonesia Sept 2023 – Dec 2023
Research Assistant
Task and Responsibility : • Contributed to “Immersive Study on People with Disabilities in Indigenous Communities”, supporting UN Indonesia’s Leave No One Behind (LNOB) initiative
• Collected secondary data on food security indices, locations of indigenous peoples, disaster risks, health conditions, and social vulnerabilities in Sumba and Jayawijaya
• Conducted and transcribed in-depth interviews with academics, CSOs, disability organizations, and disability service centers to enrich research insights